

Neeraj Kumar Pola

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Professional Summary

I build machine learning systems that move beyond experiments and deliver real impact. From improving multilingual ASR accuracy on low-resource languages to deploying production NLP pipelines that cut processing time by 12×, my work sits at the intersection of research depth and production engineering. I thrive in turning complex data and models into scalable, decision-driving systems.

Education

University at Buffalo, Masters in Artificial Intelligence. **GPA: 3.4/4** Aug 2024 – Present

• **Coursework:** Deep Learning, Machine Learning, Data Intensive Computing, RL, Algorithms, etc.

Vignana Bharathi Institute of Technology, Bachelor of Technology in Computer Science Sept 2020 – Apr 2024
Engineering (AI and ML) **GPA: 8.38/10**

• **Coursework:** Data Structures and Algorithms, Cloud Architecture, Computer Vision, NLP, etc.

Professional Experience

Machine Learning Engineer Intern, Media Sales Plus Aug 2025 – Dec 2025

- Designed and deployed a production NLP pipeline using spaCy NER, rule-based weak supervision, regex processing, and entity normalization to transform text with full correction traceability.
- Implemented end-to-end ML data pipelines with automated validation, profanity detection, error handling, and, increasing processing accuracy from 95 percent to 98 percent and enabling scalable analytics.
- Automated editorial workflows, cutting processing time from 60 minutes to 5 minutes (12× speedup), and deployed the system as a Flask service on Azure App Service for scalable, low-latency production use.

Artificial Intelligence Intern, Center for Artificial Intelligence and Robotics, DRDO Oct 2023 – Mar 2024

- Fine-tuned Wav2Vec2 transformer ASR models on 45+ hours of Bengali and Assamese audio, achieving WER 38.7 and CER 10.2 on Bengali and delivering 12–18 percent WER improvement on Assamese.
- Engineered a custom 512-unit neural layer for BPE transfer learning on 27 hours of curated Assamese audio, improving contextual representation learning on resource-scarce datasets.

Machine learning Intern, Feyn Labs Mar 2023 – May 2023

- Built forecasting models using Python, SQL, and Prophet, improving demand prediction accuracy by 15 percent.
- Performed statistical analysis, feature engineering, and customer segmentation to support data-driven decisions.

Projects

Automatic Speech Recognition(ASR) For customized language

- Wav2Vec2-based multilingual ASR system with retainable fine-tuning modules
- Built customizable Wav2Vec2 pipelines to convert audio to text for underrepresented languages.

MindMate – AI-Powered Mental Health Screening and Support

- Developed an end-to-end RAG platform integrating TiDB vector DB, LangChain orchestration, and OpenAI APIs for contextual PHQ-9 and GAD-7 assessments.
- Deployed scalable backend using FastAPI and vector search for personalized, context-aware LLM responses, supporting thousands of user sessions.

Surya Namaskar with real-time feedback using Media-pipe

- Users postures are captured in a defined sequential order, where each pose is classified and appropriate feedback is returned with high accuracy instantly.
- **Publication:** <https://www.jetir.org/view?paper=JETIR2404467>

Time Series Analysis and Prediction

- Preprocessed and analyzed 10,000+ time series records, applying feature engineering (seasonality, lags, trends).
- Built ETL pipelines feeding Prophet and LSTM models, achieving 15 percent improved forecast accuracy.

Technical Skills

Languages: Python, SQL, Java, C, R, JavaScript.

ML and Data: PyTorch, TensorFlow, Keras, Scikit-learn, NLTK, SpaCy, LangChain, Prophet, FastAPI, Spark, Airflow, Tableau, PowerBI, Generative AI, Agentic AI, LLM, RAG.

Cloud and DevOps: AWS, GCP, Azure, Docker, Kubernetes, CI/CD, Snowflake, MongoDB, MySQL.

Core Areas: Computer Vision, NLP, Time Series Forecasting, RAG Systems, Model Deployment, Statistical Modeling.

Certification: AWS Certified Cloud Practitioner.